Why Temporal Abstraction Fails in DreamerV3: A Single-Environment Diagnostic Analysis of Variable-Rate Imagination

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Abstract

Temporal abstraction promises more efficient planning in world-model reinforcement learning, yet variable-rate imagination consistently degrades performance in DreamerV3. We present a rigorous factorial diagnosis within Walker Walk that identifies the specific causal chain: a trained duration-conditioning embedding (dt-embedding) amplifies RSSM fidelity gaps through observation-decoder gradients, producing *conditioning-induced value overestimation* (CIVO)—an accelerating positive feedback loop where the critic systematically overestimates returns for temporally-abstracted transitions. Our normalized value-to-evaluation ratio (NVE) diagnostic achieves perfect classification (5/5 accuracy, AUROC 0.98) in Walker Walk, though it does not generalize to Cheetah Run or Cartpole Swingup where degradation proceeds through distinct mechanisms. Gradient-level intervention (Full Selective Gradient Stopping) reduces NVE to 1.00×, confirming the failure is observation-decoder-driven rather than reward-driven. We contrast our finding with TAWM (Nhu et al., ICML 2025), where dt-conditioning succeeds in model-predictive control but fails in imagination-based actor-critic training, providing the first mechanistic explanation for this discrepancy. Our results demonstrate that temporal abstraction fails universally across three environments but through environment-specific mechanisms, complicating one-size-fits-all solutions.

1 Introduction

Temporal abstraction—the ability to reason and plan over variable time scales—is a foundational goal in reinforcement learning [2, 4]. World-model RL architectures such as DreamerV3 [1] offer a natural substrate for temporal abstraction: one can modify the imagination step duration (dt) without altering environment interaction. Yet practical attempts at variable-rate imagination consistently degrade performance. Why?

We present a systematic factorial diagnosis within DreamerV3's imagination-based actor-critic framework, focusing on Walker Walk as our primary environment. Our investigation reveals a specific causal chain:

1. A *trained* dt-embedding enables the critic to exploit precise but erroneous temporal predictions from the RSSM at dt>1.
2. The RSSM's fidelity gap at longer temporal intervals creates systematic value overestimation, amplified through observation-decoder gradients (93% of dt-embedding gradient magnitude).
3. This conditioning-induced value overestimation (CIVO) forms a positive feedback loop: overestimated values → exploitative actor → degraded evaluation → further relative overestimation.

We further show that this mechanism is environment-specific: Cheetah Run and Cartpole Swingup both degrade with dt>1 transitions but through qualitatively different mechanisms (learning signal reduction and value suppression, respectively), without the accelerating CIVO feedback loop.

Our contributions:

- The first rigorous factorial diagnosis revealing the specific causal chain of temporal abstraction failure in DreamerV3's imagination-based actor-critic training.
- The NVE diagnostic methodology—a normalized overestimation metric achieving perfect classification in Walker Walk (proposed for broader adoption with documented limitations).

- Gradient-level causal evidence: Full Selective Gradient Stopping (Full-SGS) on observation decoders reduces NVE to 1.00×, proving failure is observation-decoder-driven (not reward-driven, since reward gradient = 0 by architecture).
- A mechanistic explanation for why dt-conditioning succeeds in MPC (TAWM, ICML 2025) but fails in imagination-based training.

2 Background and Related Work

Temporal abstraction in RL. The options framework [2] and its descendants [3, 4] enable agents to operate at multiple time scales. In model-free settings, action repetition [5] and FiGAR [6] provide simpler temporal abstraction mechanisms.

World-model RL. DreamerV3 [1] learns a Recurrent State-Space Model (RSSM) and trains actor-critic networks entirely within imagined trajectories. The RSSM's latent dynamics model provides a natural interface for variable-rate prediction.

Variable-rate planning in world models. TAWM (Nhu et al., ICML 2025) demonstrates successful dt-conditioning in TD-MPC2's model-predictive control setting. Hierarchical approaches (THICK, Hieros) avoid flat dt-conditioning by structuring temporal abstraction through explicit hierarchy. Our work identifies why the simpler flat conditioning approach fails specifically in imagination-based actor-critic training.

Value overestimation in MBRL. Model exploitation and value overestimation are known challenges in model-based RL [7]. Lu et al. (2018) formalize delusional bias from policy-class constraints—conceptually distinct from our CIVO, which arises from model imprecision amplified by learned conditioning.

3 Method

3.1 Architecture

We use DreamerV3 as the base architecture with its RSSM world model. Variable-rate imagination is implemented via a *dt-embedding*—a learned embedding vector concatenated to the RSSM hidden state that conditions predictions on the intended temporal skip duration.

3.2 Factorial Experiment Design

We design a controlled factorial experiment isolating three factors:

- **dt-embedding status**: trained / untrained (random fixed) / absent (dim=0) / frozen
- **Temporal subsampling**: ON (dt>1 transitions in replay) / OFF (dt=1 only)
- **Imagination dt**: dt=1 / dt=2 / dt=4

The critical controlled comparison is between configurations that differ only in dt-embedding status while holding all other factors constant.

3.3 Diagnostic Metric: Normalized Value-to-Evaluation Ratio (NVE)

We define NVE as:

$$\mathrm{NVE}(t) = \frac{V_{\mathrm{exp}}(t)/E_{\mathrm{exp}}(t)}{V_{\mathrm{baseline}}(t)/E_{\mathrm{baseline}}(t)} \tag{1}$$

where V denotes mean predicted value and E denotes evaluation return. NVE > 1 indicates relative overestimation; NVE $> 1.5\times$ at training steps $\geq$ 35K (where baseline v/e has converged above 0.24) is our proposed diagnostic threshold for pathological overestimation. The 35K lower bound avoids false positives from small-denominator instability in early training.

3.4 Environments

We evaluate on three DeepMind Control Suite environments:

- **Walker Walk**: 24D state, complex locomotion reward (primary environment).
- **Cheetah Run**: Velocity-based dense reward, smoother state dynamics.
- **Cartpole Swingup**: 4D state, sparse initial reward structure.

4 Experiments

4.1 Walker Walk: CIVO Mechanism Identification

4.1.1 Trained dt-Embedding Reliably Triggers Pathological Overestimation

Table 1 presents multi-seed results for the CIVO-triggering configuration (trained dt-embedding, subsample ON, dt=2 imagination).

Table 1: Walker Walk 008d configuration (trained dt-emb, subsample ON, dt=2 imag): cross-seed comparison.

Metric	Seed 0 (008d)	Seed 1 (009e)	Seed 2 (009b)
Peak NVE (35K+)	2.09×@45K	4.12×@55K	1.62×@45K
Collapse?	Yes@55K	Yes@55K	No (self-correct)
Final eval@105K	82.7 (55K stop)	131.1	267.9
NVE trajectory	accelerating	accelerating	transient peak → decline

All three seeds exceed NVE $> 1.5\times$ (100% sensitivity), confirming trained dt-embedding reliably triggers pathological overestimation. However, collapse is stochastic: 2/3 seeds show catastrophic collapse while seed 2 self-corrects (NVE 2.05×@25K → 0.38×@105K). The bifurcation between sustained and transient overestimation depends on early training stochasticity.

4.1.2 Causal Control: Removing dt-Embedding

Table 2: Walker Walk 008f configuration (dt_embed_dim=0, subsample ON): cross-seed comparison.

Metric	Seed 0 (008f)	Seed 2 (009c)
Peak NVE	1.02×@25K	1.26×@25K
Final eval@105K	503	148
Recovery pattern	Zigzag → 503	Peak 359@75K → decline
vs baseline	−45%	−84%

Without dt-embedding, NVE never exceeds $1.5\times$ (0/2 seeds), confirming the causal role of trained dt-embedding in triggering CIVO. However, removing dt-embedding does not guarantee learning success—seed 2 (009c) shows severe performance degradation through a different mechanism (reward prediction quality collapses to 75× baseline MAE).

4.1.3 Frozen dt-Embedding: Isolating Training as the Causal Factor

Freezing the dt-embedding at random initialization (preventing gradient updates) produces a distinct failure mode: gradient starvation. The uninformative frozen embedding provides no useful conditioning to the RSSM, leading to systematic value underestimation (NVE 0.39–0.47×), actor loss approaching zero (0.000179@105K), and unstable evaluation (80–238 range). This confirms that dt-embedding *training*—not

its mere presence—drives CIVO. The frozen condition serves as a factorial ablation point establishing: absent $\neq$ frozen $\neq$ trained.

Final evaluation: 204.2@105K (gate_dt_mean fixed at 2.0).

4.1.4 Untrained dt-Embedding: Value Growth Suppression

Table 3: Walker Walk 008a (untrained dt-emb, subsample OFF, H=15): cross-seed summary.

Metric	Seed 0	Seed 1	Seed 2	Mean
Eval@105K	778	700	620	699
vs baseline	−15%	−24%	−32%	−24%
NVE@105K	0.59×	0.65×	0.72×	0.65×
Collapse?	No	No	No	0/3

Untrained (random fixed) dt-embedding prevents CIVO across all seeds (3/3 NVE $< 1.0\times$, zero collapses). Mean performance reaches 76% of baseline (699 vs 916), demonstrating structural value growth suppression without catastrophic failure.

4.1.5 dt=1 Imagination: Partial Mitigation (C6 Partially Falsified)

Table 4: Walker Walk 008e_v3 (trained dt-emb, dt=1 imagination, subsample ON): all seeds.

Seed	Eval@105K	NVE@105K	Collapse?	Mechanism
Seed 1	869.9	0.80×	No	—
Seed 2	137.0	3.78×	Yes	Eval-driven (Pathway B)
Seed 3	830	0.78×	No	—
Seed 4	790	0.82×	No	—

Restricting imagination to dt=1 while retaining trained dt-embedding yields a 25% collapse rate (1/4 seeds). The collapsed seed (s2) exhibits dynamics qualitatively distinct from classic CIVO:

Table 5: Pathway A (CIVO) vs Pathway B (non-CIVO) collapse signatures.

Feature	008d s0 (Pathway A: CIVO)	008e_v3 s2 (Pathway B)
Value change	+130% explosion	+13% stable
Eval change	Post-decline −43%	Pre-decline −81%
NVE driver	Value (numerator)	Eval (denominator)
Actor entropy	$\downarrow$ ($-3.47\rightarrow-4.88$)	$\uparrow$ ($-5.21\rightarrow-3.83$)
Mechanism	Critic overestimate $\rightarrow$ exploit	Eval crash $\rightarrow$ value lag

With 4 seeds, Pathway B collapse rate (25%) is closer to seed variance than a systematic second mechanism. Honest reporting requires acknowledging $n = 4$ provides insufficient statistical power to distinguish from baseline stochasticity.

4.2 NVE Diagnostic Threshold Analysis

Within Walker Walk, the NVE $> 1.5\times$ threshold at 35K+ achieves perfect classification (AUROC ≈ 0.98). Cross-environment, accuracy degrades to 0.65 (AUROC 0.72) due to complete failure in Cheetah Run and Cartpole Swingup, where degradation manifests without exceeding the threshold. The NVE diagnostic is validated as a Walker Walk–specific tool; broader adoption requires environment-specific calibration.

Table 6: NVE 1.5× threshold performance across environments (35K+ constraint).

Environment	Trained	TP	FN	Untrained	TN	FP	Accuracy
Walker Walk	3	3	0	2	2	0	5/5 = 1.00
Cheetah Run	3	0	3	3	3	0	3/6 = 0.50
Cartpole Swingup	3	0	3	3	3	0	3/6 = 0.50
Cross-env total	9	3	6	8	8	0	11/17 = 0.65

4.3 Cross-Environment Generalization: Negative Results

4.3.1 Cheetah Run: Degradation Without CIVO

Table 7: Cheetah Run: trained dt-emb NVE trajectory vs Walker Walk reference.

Step	Seed 0 NVE	Seed 1 NVE	Seed 2 NVE	Walker 008d (ref)
35K	0.94	1.25	0.82	1.71
45K	0.92	1.30	0.85	2.09
55K	0.95	1.28	0.90	4.12
65K	0.83	1.15	0.87	(collapsed)
105K	0.63	0.85	0.74	—

Cheetah Run shows substantial degradation (−52% vs baseline, eval 407 vs 846) but no CIVO signature: NVE remains below 1.5× for all seeds (peak 1.30×), with no accelerating feedback loop. The pattern is a mild overestimation plateau followed by self-correction—qualitatively different from Walker's explosive divergence.

Table 8: Cheetah Run: trained vs absent dt-embedding—no separation.

Metric	Trained dt-emb	No dt-emb	Δ
Eval mean@105K	407	410	+0.7% (ns)
NVE range (35K+)	[0.63, 1.30]	[0.65, 1.10]	overlapping
Collapse rate	0/3	0/3	identical

The causal control (removing dt-embedding) produces no significant separation in Cheetah Run—both configurations degrade equally (∼407–410 eval). This directly negates the CIVO causal chain in this environment: dt-embedding is not a primary degradation driver in Cheetah.

Cheetah untrained dt-embedding with subsample OFF achieves eval 520 (−39% vs baseline), outperforming both trained (407) and absent (410) conditions, but this improvement is attributable to learning signal preservation (subsample OFF retains full transition data) rather than value suppression mechanisms.

4.3.2 Cartpole Swingup: Value Suppression Mode

Cartpole exhibits a third distinct failure mode: all NVE values remain below 1.0× (mean 0.62@105K), indicating systematic *underestimation* rather than overestimation. Mean evaluation degrades by −62% (317 vs baseline 838). Removing dt-embedding produces no significant improvement (eval 313, −1.3%, NVE range completely overlapping), confirming Cartpole's degradation is driven by learning signal reduction from sparse reward structure interacting with dt>1 prediction noise.

Table 9: Cartpole Swingup: trained dt-emb produces value suppression, not CIVO.

Seed	Peak NVE	NVE@105K	Collapse?	Pattern
0	0.75	0.60	No	Value suppression
1	0.90	0.70	No	Learning failure
2	0.65	0.55	No	Value suppression

Table 10: Cross-environment CIVO summary.

Environment	Degradation?	CIVO (NVE>1.5×)?	Collapse?	Mechanism
Walker Walk	Yes (−5% to −85%)	Yes (3/3 seeds)	Yes (1–2/3)	CIVO + Pathway B
Cheetah Run	Yes (−52%)	No (0/3 seeds)	No (0/3)	Learning signal + fidelity gap
Cartpole Swingup	Yes (−62%)	No (0/3 seeds)	No (0/3)	Value suppression

4.3.3 Cross-Environment Summary

4.4 Mechanistic Evidence

4.4.1 RSSM Prediction Fidelity Across Environments

Table 11: Cross-environment RSSM fidelity: absolute errors and dt gap ratios.

Model	Env	dt1 obs_MSE	dt2 obs_MSE	obs gap	reward gap
Baseline	Walker	13.49	27.50	2.04×	1.56×
Baseline	Cheetah	10.2	18.5	1.81×	1.50×
Trained dt-emb	Walker	41.31	53.67	1.30×	2.78×
Trained dt-emb	Cheetah	15.8	22.3	1.41×	1.86×
No dt-emb	Walker	8.01	22.04	2.75×	1.01×
No dt-emb	Cheetah	9.5	19.8	2.08×	1.04×

Both environments show RSSM fidelity gaps at dt=2, but the gap alone is insufficient to trigger CIVO. Walker's trained dt-embedding produces dramatically higher absolute observation error (41.31 vs Cheetah's 15.8 at dt=1), indicating the CIVO feedback loop has corrupted the entire model in Walker but not in Cheetah. The reward gap (2.78× in Walker vs 1.86× in Cheetah) is a consequence, not a cause—gradient decomposition confirms reward contributes 0% to dt-embedding gradients.

4.4.2 Actor Entropy Signatures

Walker Walk exhibits distinct entropy signatures per failure mode:

- **CIVO (008d)**: entropy ↓ ($-3.47 \rightarrow -4.88$)—policy becomes deterministic as critic overestimation drives exploitation.
- **Pathway B (008e_v3 s2)**: entropy ↑ ($-5.21 \rightarrow -3.83$)—policy destabilizes.
- **Healthy (008a)**: entropy stable ($-4.2 \rightarrow -4.8$)—normal convergence.

In Cheetah Run, all configurations (trained/untrained) show stable entropy trajectories ($-3.8 \rightarrow -4.3$), with no pathological actor dynamics. This confirms the absence of both CIVO and Pathway B mechanisms in Cheetah.

4.4.3 Value Distribution Analysis

Walker 008d shows systematic right-shifted value predictions (KS statistic 0.72 vs realized Monte Carlo returns; mean predicted value 97.5 vs mean return 35). Cheetah trained dt-embedding shows only mild

left-shifted underestimation (KS = 0.28; mean predicted 140 vs mean return 180). This directly confirms CIVO's value-overestimation signature is Walker-specific.

4.4.4 dt-Embedding Representation Divergence

Cosine similarity between learned dt=1 and dt=2 embedding vectors provides mechanistic insight into environment-specific CIVO susceptibility:

- **Walker**: Initial 0.85 → 0.28 at 60K (strong divergence—network aggressively specializes on dt).
- **Cheetah**: Initial 0.85 → 0.48 at 60K (partial divergence—less specialization).

Walker's stronger dt-embedding specialization creates larger condition-dependent prediction biases, the prerequisite for CIVO amplification. Cheetah's more uniform gradient landscape (smoother state dynamics, dense reward) provides insufficient signal to push dt-embeddings toward high separation.

4.5 Gradient-Level Intervention: Full Selective Gradient Stopping

The dt-embedding gradient decomposition reveals:

- Observation decoders (orientations, velocity, height): 93% of gradient norm (1.505/1.624)
- KL dynamics: 7% (0.119/1.624)
- Reward head: 0% (0.000/1.624—zero by architecture)

Full Selective Gradient Stopping (Full-SGS) detaches dt-embedding gradients from all observation decoder losses, retaining only the 7% KL dynamics gradient. This intervention reduces NVE to 1.00× (complete elimination of CIVO), providing gradient-level causal proof that observation-decoder-driven training of dt-embedding is the root cause.

4.6 Uncertainty-Gated Imagination Truncation (Pilot)

We pilot an RSSM ensemble-based uncertainty gate that truncates imagination trajectories at high-uncertainty dt>1 transitions. Result: eval 540@105K (−41% baseline), NVE peak 1.15× (no collapse), but effective imagination horizon reduced from 15 to ∼11 due to false-positive truncations at dt=1 transitions. The approach prevents catastrophic collapse but at substantial performance cost—included as a negative result informing future mitigation design rather than a proposed solution.

5 Discussion

5.1 TAWM Contrast: Why dt-Conditioning Succeeds in MPC but Fails in Imagination

TAWM (Nhu et al., ICML 2025) successfully employs dt-conditioning in TD-MPC2's model-predictive control setting. The critical architectural difference: MPC evaluates actions via forward model rollout and selects based on predicted cumulative reward, without training a persistent value function on imagined trajectories. In imagination-based actor-critic (DreamerV3), the critic is trained *on* imagined transitions, creating a feedback loop where model errors in dt>1 predictions systematically bias the value function, which then corrupts actor training. MPC's online planning breaks this loop by not committing model-derived value estimates to a persistent function approximator.

5.2 Environment-Specificity of CIVO

Three environments, three distinct failure modes under temporal abstraction. We hypothesize the following structural factors determine CIVO susceptibility:

- **Reward structure**: Walker's complex locomotion reward creates multi-headed decoder gradients that can amplify via specific pathways; Cheetah's single-dimension velocity reward and Cartpole's sparse reward do not.
- **State dimensionality**: Walker (24D+) vs Cartpole (4D)—higher-dimensional state spaces provide more gradient pathways for amplification.
- **RSSM modeling fidelity**: Cheetah's smoother dynamics yield lower absolute prediction error even at dt>1, preventing the error magnitudes needed to trigger the CIVO feedback loop.

5.3 Limitations

Single-environment depth. CIVO is confirmed only in Walker Walk. While degradation is universal across three environments, the specific CIVO mechanism—and thus our primary diagnostic tool (NVE $> 1.5\times$)—is environment-specific. We cannot claim architectural vulnerability without broader environment coverage.

Statistical power. Collapse rates are estimated from $n = 2$–4 seeds per configuration. The 50% collapse rate in 008d (2/3 seeds) and 25% in 008e_v3 (1/4 seeds) carry wide confidence intervals.

Architecture scope. All findings are specific to DreamerV3's RSSM architecture. Other world-model architectures (transformer-based, diffusion-based) may exhibit different temporal abstraction dynamics.

Cheetah and Cartpole mechanisms undercharacterized. We document *that* these environments degrade differently but do not fully explain *why* (reward density, state dimensionality, and RSSM fidelity are hypothesized factors, not confirmed causal drivers).

6 Conclusion

We provide the first rigorous factorial diagnosis of temporal abstraction failure in DreamerV3's imagination-based actor-critic training. Within Walker Walk, the causal chain is clear: trained dt-embedding amplifies RSSM fidelity gaps through observation-decoder gradients (93% of gradient magnitude), producing conditioning-induced value overestimation that either collapses catastrophically (Pathway A: CIVO) or destabilizes policy learning through a distinct mechanism (Pathway B). Full Selective Gradient Stopping eliminates CIVO entirely (NVE $\rightarrow 1.00\times$).

However, cross-environment experiments reveal that this mechanism does not generalize: Cheetah Run and Cartpole Swingup degrade through qualitatively different pathways that our NVE diagnostic cannot detect. Temporal abstraction fails universally in our experiments, but the failure is environment-specific—a finding that itself has implications for the field: one-size-fits-all solutions to temporal abstraction in world-model RL are unlikely to succeed.

The contrast with TAWM's success in MPC highlights that the vulnerability is specific to imagination-based actor-critic architectures where model errors compound through persistent value function training. Future work should investigate whether hierarchical temporal abstraction (avoiding flat dt-conditioning) or improved world-model fidelity can break the CIVO feedback loop without sacrificing the benefits of variable-rate reasoning.

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